

V3 empirical study · Instruction

Six models · three underlyings · four volatility regimes · 5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of six American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: GBM, GARCH(1,1), Heston (no jumps), Merton jump-diffusion, GARCH-Merton, Heston-Merton (Bates).
- Companies: SPY (primary), AAPL, MSFT. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 5 calendar years ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than five years, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

Six models · three underlyings · four volatility regimes · 5-year monthly calibration

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all six models: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: (κ , θ , ξ , ρ , v_0) are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 2000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean } |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all six models)

Regime	Window	SPY n	AAPL n	MSFT n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50

Filter funnel · last-step n in the 5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	$ S/K-1 \leq 10\%$	Liquidity
SPY	2008-2009	3542	3490	3490	3377	3377
SPY	2013-2014	15498	15131	15131	14935	14935
SPY	2018-2019	43198	42978	42978	42596	42596
SPY	2019-2020	68155	67933	67933	67136	67136
AAPL	2008-2009	728	728	728	691	691
AAPL	2013-2014	10915	10852	10852	10547	10547
AAPL	2018-2019	8697	8681	8681	8383	8383
AAPL	2019-2020	12668	12643	12643	12211	12211
MSFT	2008-2009	584	584	584	556	556
MSFT	2013-2014	3575	3512	3512	3438	3438
MSFT	2018-2019	10040	9910	9910	9711	9711
MSFT	2019-2020	13143	13015	13015	12698	12698

*Input is the processed call panel restricted to (regime end – 5 years, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with `trading_date` $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All six models still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-year lookback, monthly rolling, n_paths=2000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	38.42	1.4383	-1.1776	-22.34	26.9%	3
GARCH	46.23	1.3257	+1.1985	+32.55	24.0%	5
Heston	33.94	1.1172	+0.6042	+21.77	29.1%	2
Merton	39.03	1.4566	-1.2870	-25.81	20.6%	4
GARCH-Merton	46.54	1.3459	+1.2347	+33.36	24.4%	6
Heston-Merton	32.91	1.0991	+0.6089	+21.17	27.4%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	100.28	2.0653	+2.0590	+90.52	28.9%	6
GARCH	57.15	1.0690	+1.0241	+46.67	26.0%	2
Heston	49.92	0.9511	+0.8255	+37.88	31.4%	1
Merton	97.95	2.0325	+2.0214	+88.70	25.0%	5
GARCH-Merton	68.35	1.3255	+1.2907	+58.09	25.7%	4
Heston-Merton	61.00	0.9735	+0.6775	+34.01	18.5%	3

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	37.73	1.2009	+0.7363	+22.35	24.5%	2
GARCH	45.22	1.3736	+0.8722	+25.21	26.2%	3
Heston	61.87	2.0402	+1.7667	+45.91	25.2%	5
Merton	34.83	1.1179	+0.6783	+20.60	23.4%	1
GARCH-Merton	51.27	1.5665	+1.2336	+33.72	23.3%	4
Heston-Merton	71.48	2.3922	+2.1412	+54.46	23.7%	6

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	47.76	2.6358	-0.2100	+18.56	27.0%	3
GARCH	40.67	2.2735	+0.8863	+20.90	34.0%	1
Heston	68.26	3.7347	+2.8964	+56.73	30.4%	6
Merton	47.73	2.6129	-0.3988	+16.01	21.6%	2
GARCH-Merton	48.34	2.5391	+1.2739	+29.35	31.0%	4
Heston-Merton	64.54	3.6163	+2.7561	+52.39	24.8%	5

Page 5 · Table block · AAPL · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-year lookback, monthly rolling, n_paths=2000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	41.53	1.9312	+0.2387	+10.75	25.1%	5
GARCH	36.79	1.6244	+1.1309	+21.73	21.7%	3
Heston	32.43	1.6546	+1.1227	+21.96	27.1%	1
Merton	37.67	1.8257	+0.1695	+9.16	23.8%	4
GARCH-Merton	44.68	2.0042	+1.7684	+32.00	20.7%	6
Heston-Merton	34.87	1.7419	+1.2144	+23.76	24.4%	2

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	52.38	3.4797	+3.4797	+44.91	28.6%	4
GARCH	66.76	4.4660	+4.4660	+52.71	25.3%	5
Heston	38.62	2.5601	+2.5591	+30.71	26.3%	2
Merton	50.97	3.4661	+3.4661	+43.83	24.6%	3
GARCH-Merton	79.50	5.6167	+5.6167	+67.98	22.1%	6
Heston-Merton	36.09	2.3642	+2.3414	+25.14	21.0%	1

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	21.25	1.0474	+0.4108	+8.52	26.3%	2
GARCH	30.89	1.2986	+0.8518	+16.58	23.8%	3
Heston	45.30	2.3807	+2.3093	+38.77	28.5%	6
Merton	20.15	1.0021	+0.3123	+7.06	24.1%	1
GARCH-Merton	38.09	1.6654	+1.5184	+27.21	23.3%	4
Heston-Merton	42.90	2.1834	+2.0613	+34.79	26.1%	5

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	31.58	2.6323	-0.1333	+8.72	28.0%	2
GARCH	32.99	2.6019	+1.2205	+17.91	27.1%	3
Heston	42.43	3.7882	+3.0597	+33.78	29.9%	6
Merton	30.78	2.5957	-0.4305	+6.36	25.7%	1
GARCH-Merton	41.45	3.1375	+2.1560	+27.93	24.2%	5
Heston-Merton	41.18	3.6295	+2.8439	+31.59	27.3%	4

Page 6 · Table block · MSFT · six models × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 5-year lookback, monthly rolling, n_paths=2000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	31.97	0.3575	-0.2783	-18.69	27.9%	3
GARCH	39.19	0.3113	+0.1524	+19.04	26.0%	5
Heston	28.27	0.2618	+0.1014	+13.94	29.9%	1
Merton	32.44	0.3574	-0.3057	-21.58	24.0%	4
GARCH-Merton	43.26	0.3572	+0.2363	+26.88	25.4%	6
Heston-Merton	28.31	0.2706	+0.1179	+15.29	25.6%	2

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	61.21	0.4308	+0.4287	+53.07	28.1%	5
GARCH	59.89	0.4075	+0.4025	+50.11	25.0%	3
Heston	32.71	0.1999	+0.1659	+22.40	27.2%	2
Merton	60.82	0.4214	+0.4179	+52.16	25.4%	4
GARCH-Merton	75.98	0.5473	+0.5473	+66.89	24.8%	6
Heston-Merton	31.20	0.1950	+0.1452	+20.06	21.8%	1

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	32.73	0.8125	+0.5467	+19.50	26.1%	2
GARCH	41.85	1.0443	+0.9617	+31.75	25.0%	5
Heston	37.65	0.9636	+0.8374	+26.73	28.7%	4
Merton	31.36	0.7856	+0.4968	+17.86	22.2%	1
GARCH-Merton	53.56	1.4284	+1.4172	+45.31	21.7%	6
Heston-Merton	35.14	0.8704	+0.7218	+23.07	20.9%	3

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
GBM	40.18	1.5437	+0.0320	+14.33	27.5%	3
GARCH	46.39	1.6872	+1.0123	+28.38	31.1%	5
Heston	41.31	2.1263	+1.5877	+31.36	28.1%	4
Merton	40.00	1.5733	-0.0614	+13.07	23.7%	2
GARCH-Merton	57.86	2.1145	+1.5977	+40.43	26.6%	6
Heston-Merton	36.07	1.8444	+1.1975	+21.37	22.5%	1

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 12 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 12 cells best
1	Heston	42.73	39.96	3
2	Heston-Merton	42.97	36.08	4
3	Merton	43.64	38.35	4
4	GBM	44.75	39.30	0
5	GARCH	45.34	43.53	1
6	GARCH-Merton	54.07	49.81	0

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	Total
GBM	0	0	0	0
GARCH	1	0	0	1
Heston	1	1	1	3
Merton	1	2	1	4
GARCH-Merton	0	0	0	0
Heston-Merton	1	1	2	4

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
GBM	37.30	71.29	30.57	39.84	44.75
GARCH	40.73	61.27	39.32	40.02	45.34
Heston	31.55	40.42	48.27	50.67	42.73
Merton	36.38	69.91	28.78	39.50	43.64
GARCH-Merton	44.83	74.61	47.64	49.22	54.07
Heston-Merton	32.03	42.76	49.84	47.27	42.97

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
GBM	56.05	43.09	0
GARCH	47.32	45.73	1
Heston	53.50	55.90	1
Merton	54.88	43.38	1
GARCH-Merton	53.63	49.81	0
Heston-Merton	57.48	62.77	1

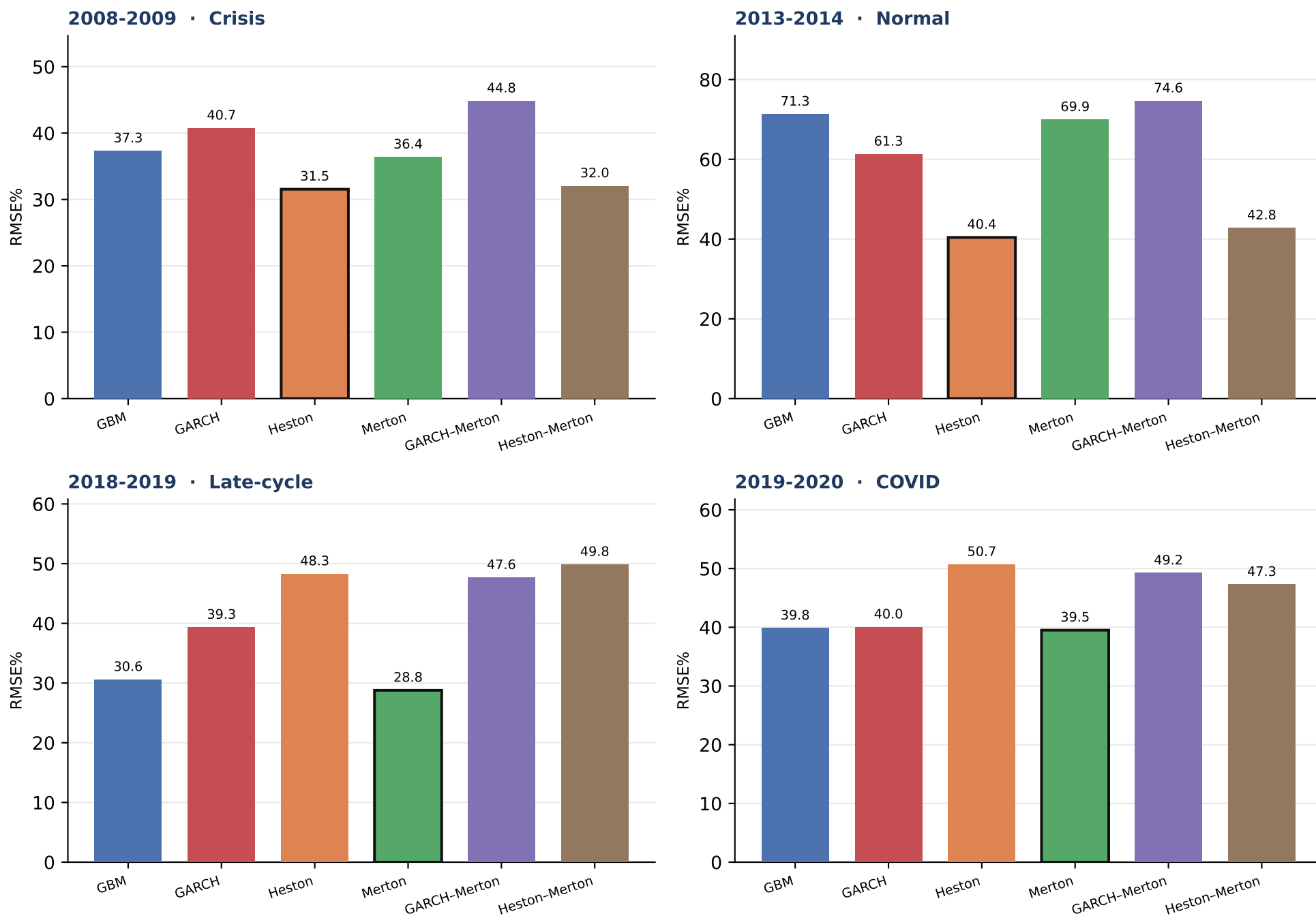
Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
GBM	36.68	36.55	0
GARCH	41.86	34.89	0
Heston	39.69	40.52	1
Merton	34.89	34.23	2
GARCH-Merton	50.93	43.06	0
Heston-Merton	38.76	38.64	1

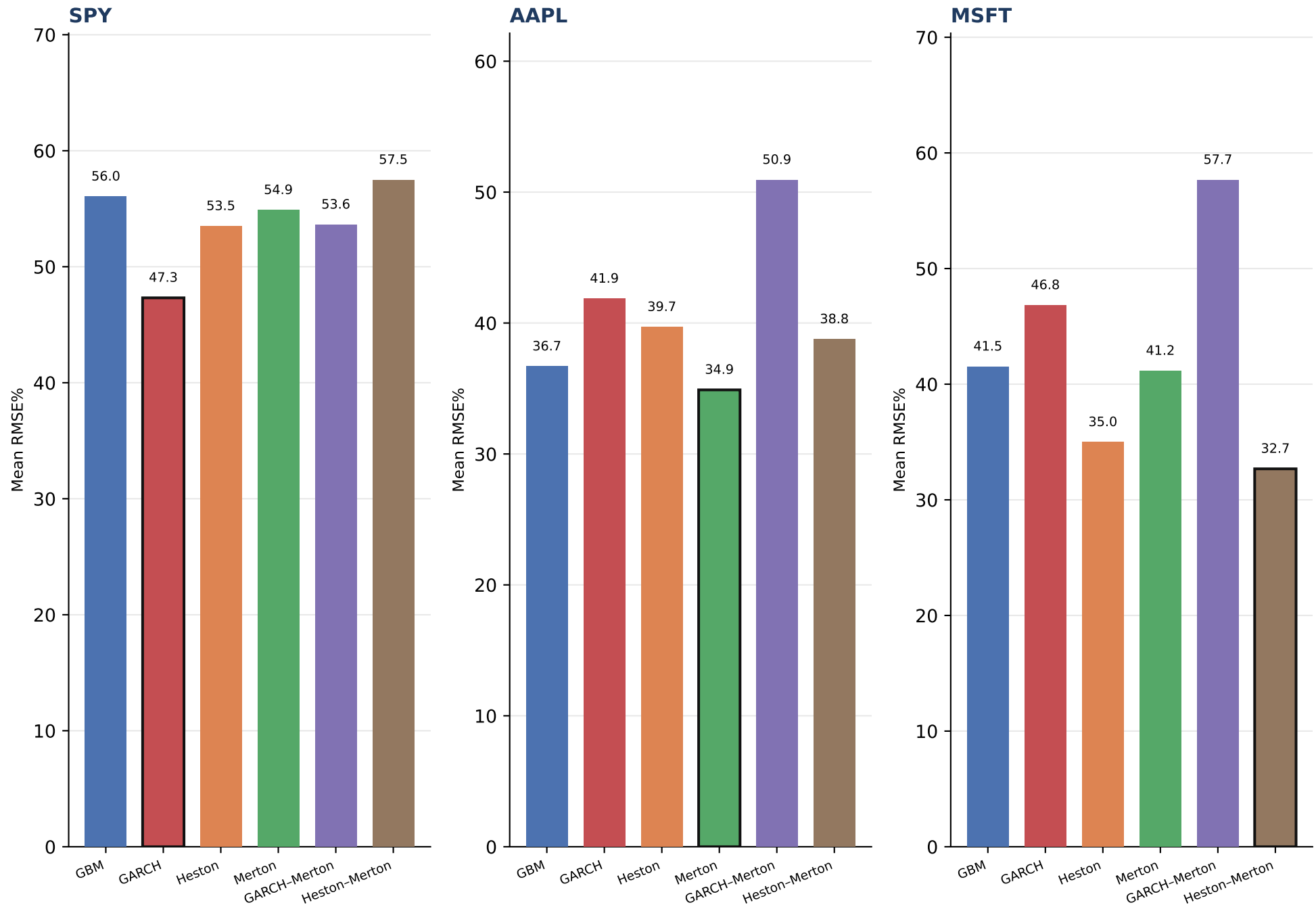
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
GBM	41.52	36.46	0
GARCH	46.83	44.12	0
Heston	34.99	35.18	1
Merton	41.15	36.22	1
GARCH-Merton	57.66	55.71	0
Heston-Merto	32.68	33.17	2

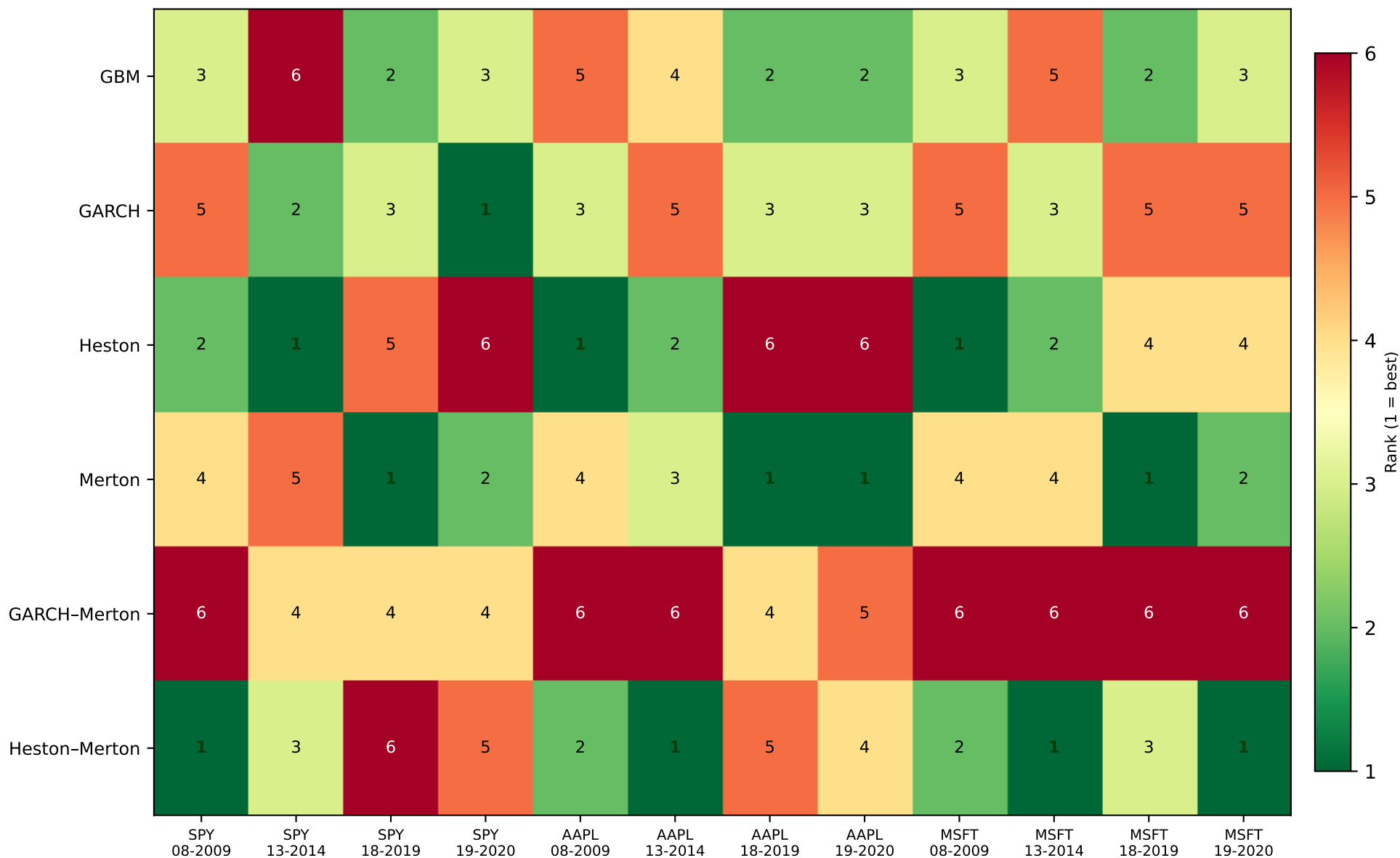
Each column is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



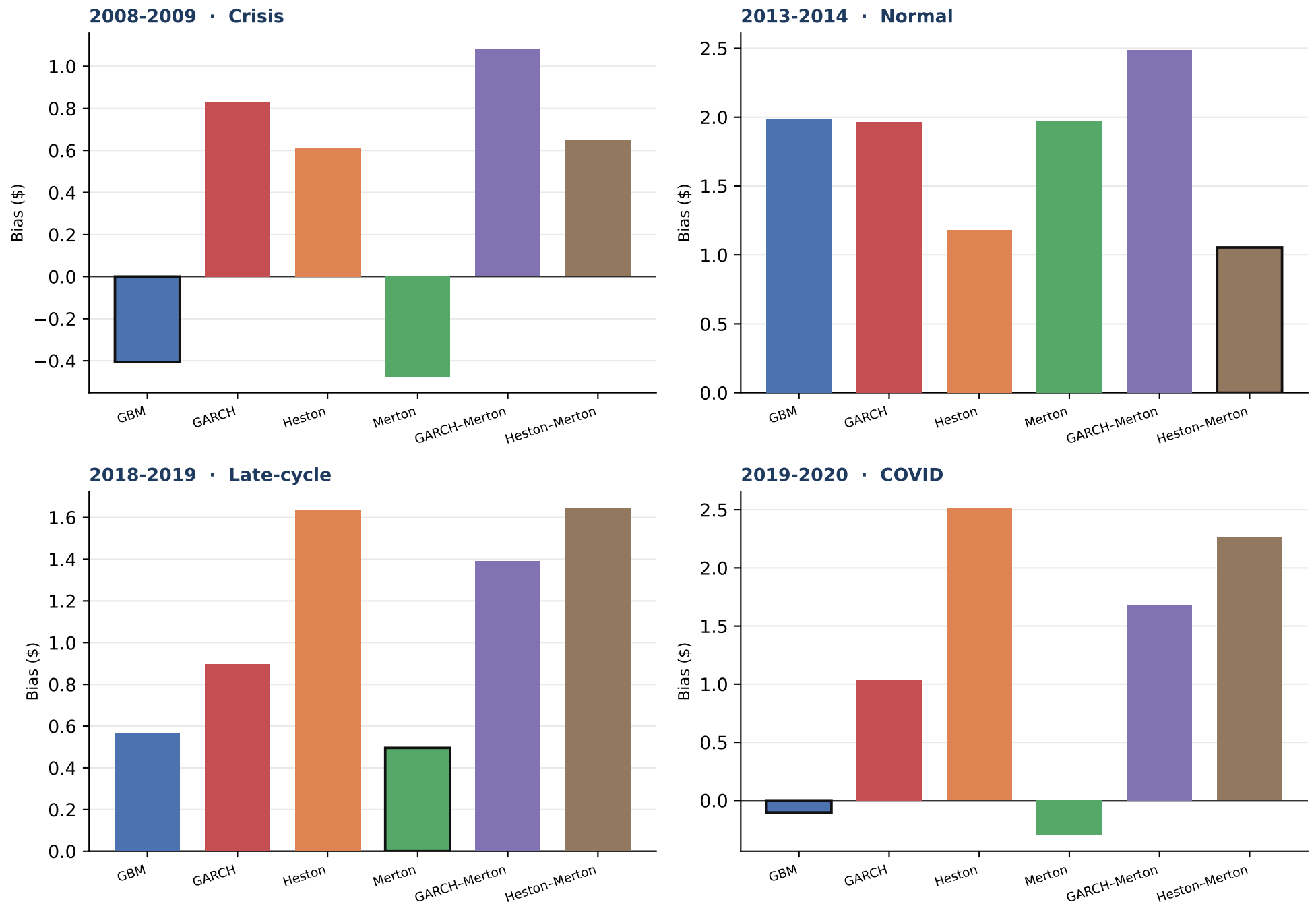
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the three companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 12 company×regime cells, Heston is the best overall model (mean RMSE% = 42.73; 3 of 12 cells).
- Second overall is Heston-Merton (mean RMSE% = 42.97; 4 cells).
- The ranking is not stable. Best-in-cell models across the 12 tables: GARCH, Heston, Heston-Merton, Merton.
- SPY: lowest mean RMSE% is GARCH (47.32); wins 1 of 4 regimes.
- AAPL: lowest mean RMSE% is Merton (34.89); wins 2 of 4 regimes.
- MSFT: lowest mean RMSE% is Heston-Merton (32.68); wins 2 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Heston (31.55).
- 2013-2014 (Normal): best mean RMSE% is Heston (40.42).
- 2018-2019 (Late-cycle): best mean RMSE% is Merton (28.78).
- 2019-2020 (COVID): best mean RMSE% is Merton (39.50).
- No-jump models as a group have lower mean RMSE% (44.27) than the three jump models (46.90).
- For the overall winner (Heston), crisis RMSE% is 31.55 vs 40.42 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 5-year monthly calibration, Monday ATM sample, LSM with 2,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all six models. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.